Software Architecture and Process Models
INTE 22293- Academic Year 2024/2025
C4 Model





University of Kelaniya, Sri Lanka 

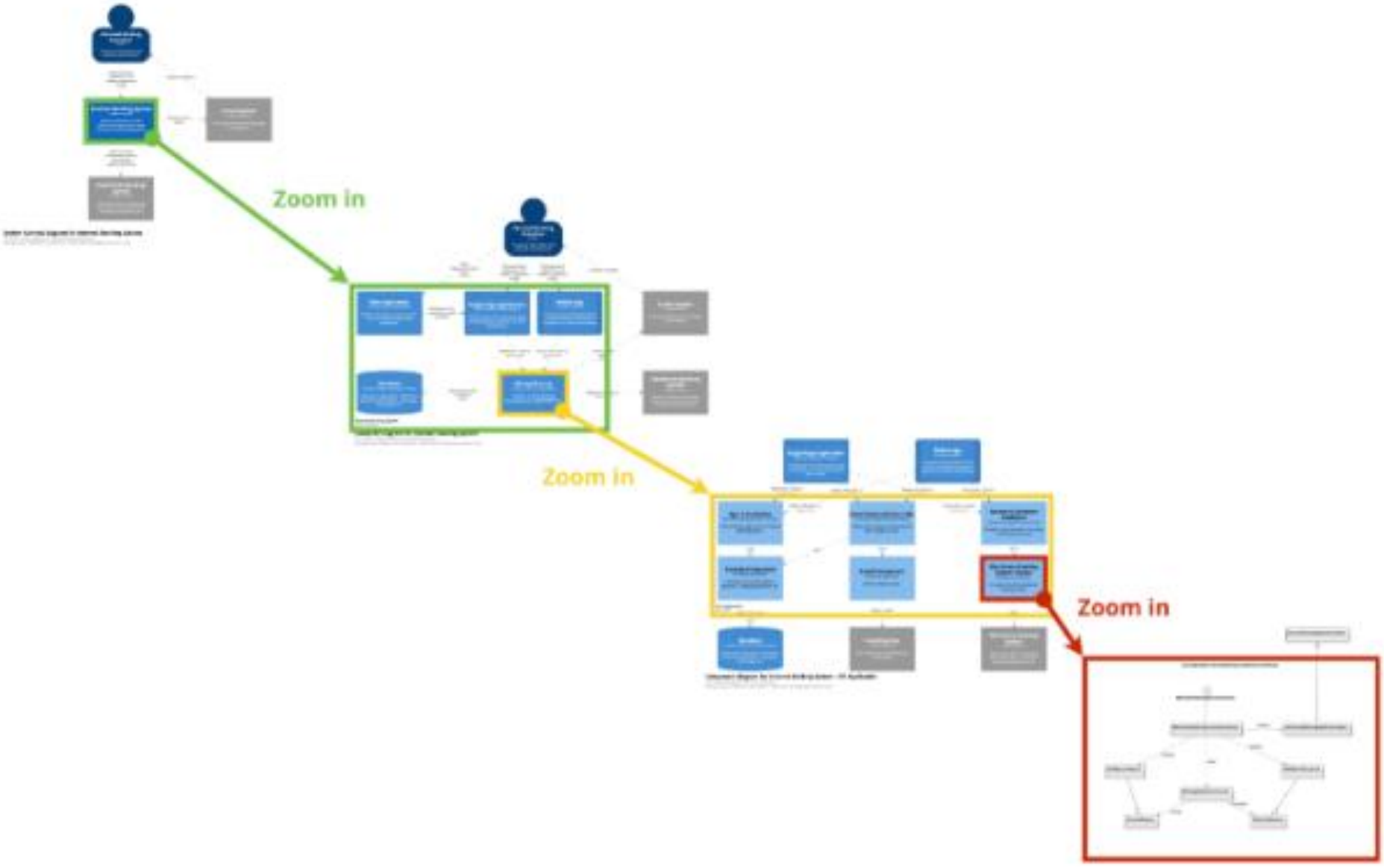
Dr. Dilani Wickramaarachchi

PhD, MSc., BSc. (Special)

Senior Lecturer

Department of Industrial Management, Faculty of Science

E-Mail – dilani@kln.ac.lk



Level 1
Context

Level 2
Containers

Level 3
Components

Level 4
Code

C4 Model

- The **C4 Model** is a lean, developer-friendly approach for visualising software architecture by breaking down a system into four hierarchical levels of detail: **Context**, **Containers**, **Components**, and **Code**.

- Simon Brown is the creator of the C4 model for visualising software architecture

- Refer c4model.com



Explanation of the C4 Model

Level 1: System Context Diagram

Shows the big picture, who uses the system (users/roles) and what external software systems it interacts with.

Level 2: Container Diagram

Zooms into the system boundary to show high-level technical building blocks (web applications, mobile apps, API services, databases) and how they communicate over protocols).

Level 3: Component Diagram

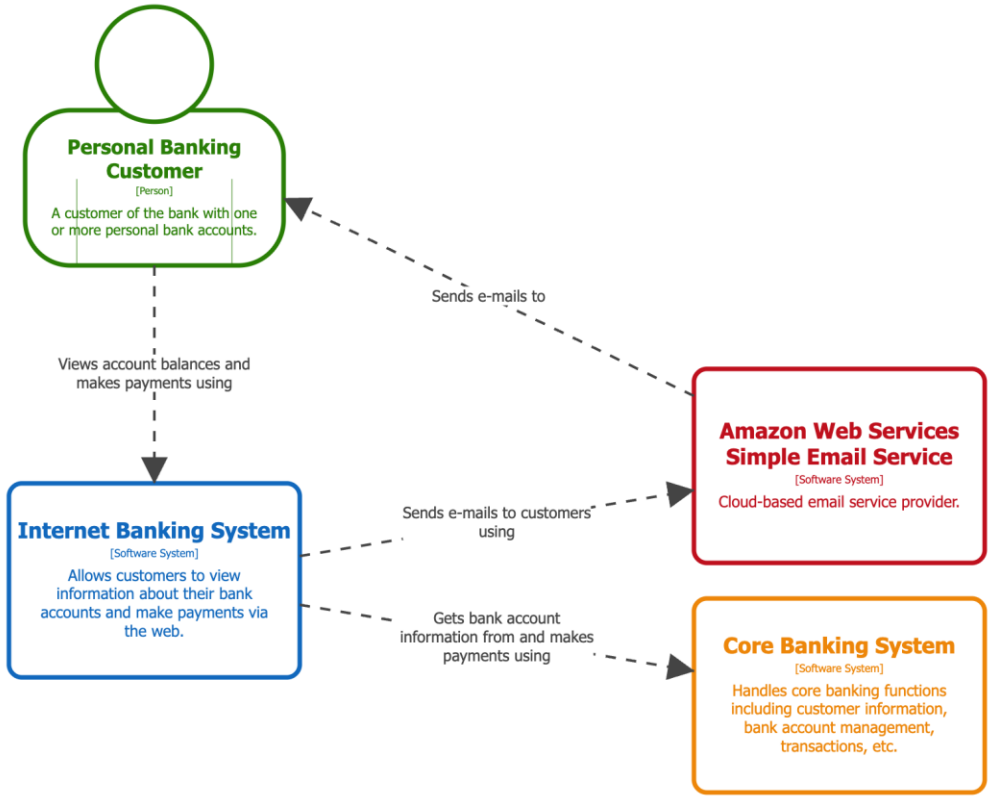
Zooms into a single container to show its structural components (controllers, repositories, authentication services) and their internal interactions.

Level 4: Code Diagram

Zooms into an individual component to show implementation details



Context Diagram

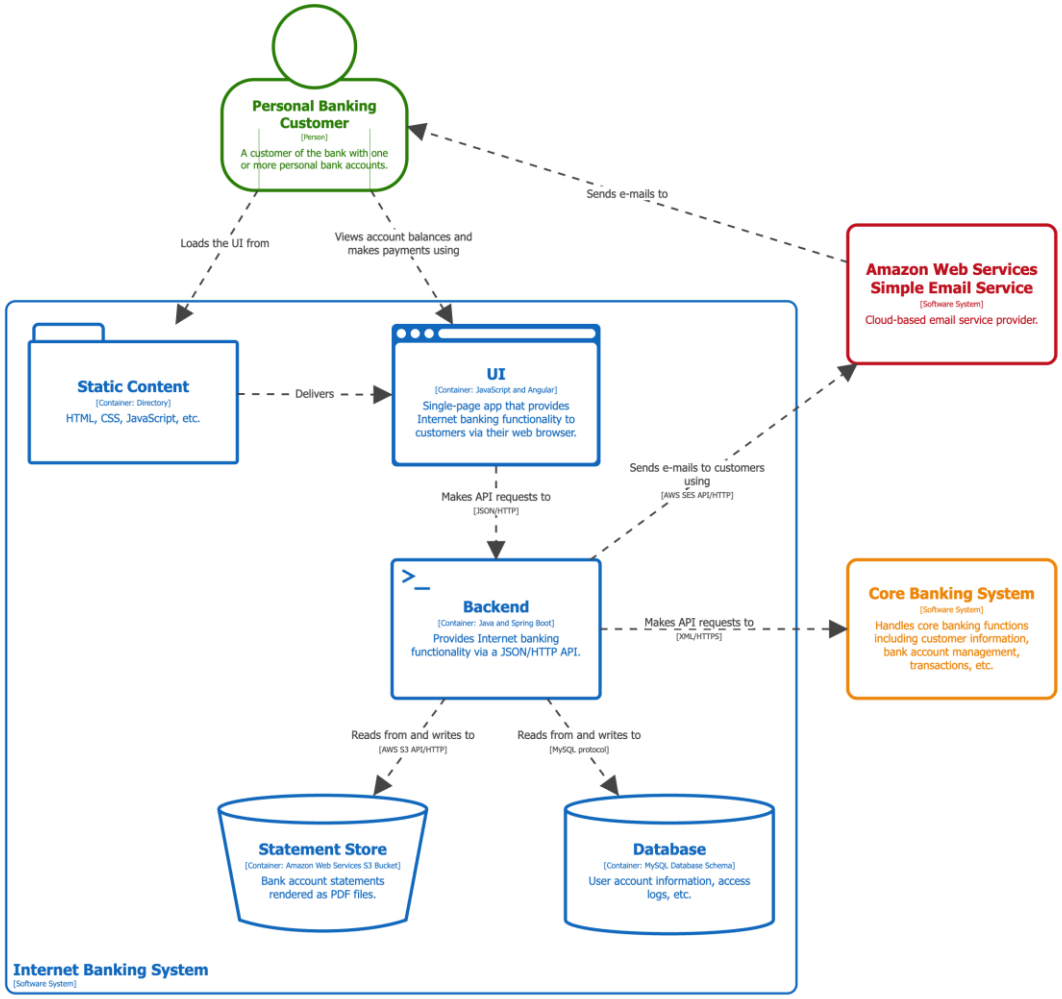


System Context View: Internet Banking System

The system context diagram for a fictional Internet Banking System | Simon Brown | c4model.com | License: CC BY 4.0



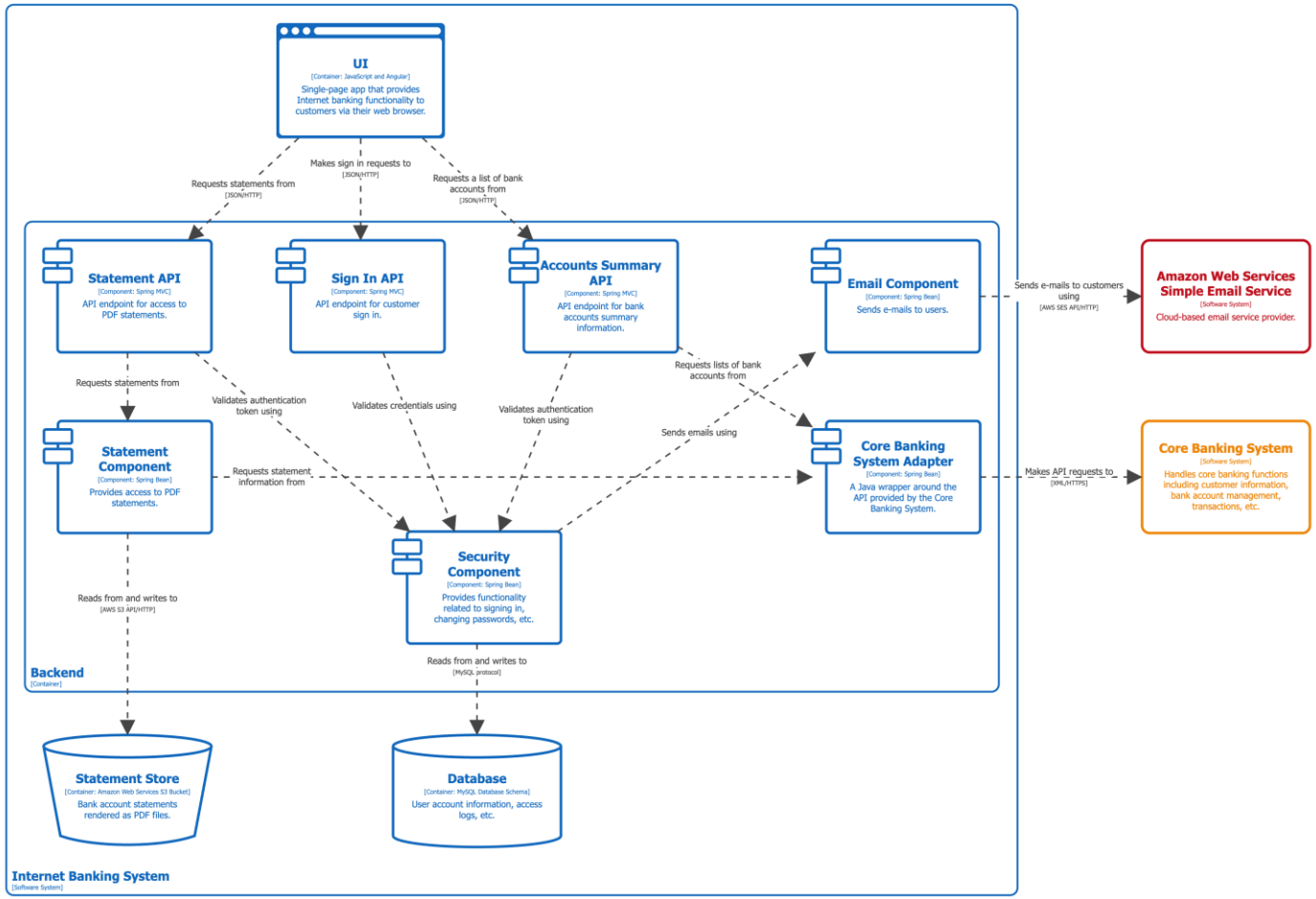
Container Diagram



Container View: Internet Banking System

The container diagram for the Internet Banking System | Simon Brown | c4model.com | License: CC BY 4.0

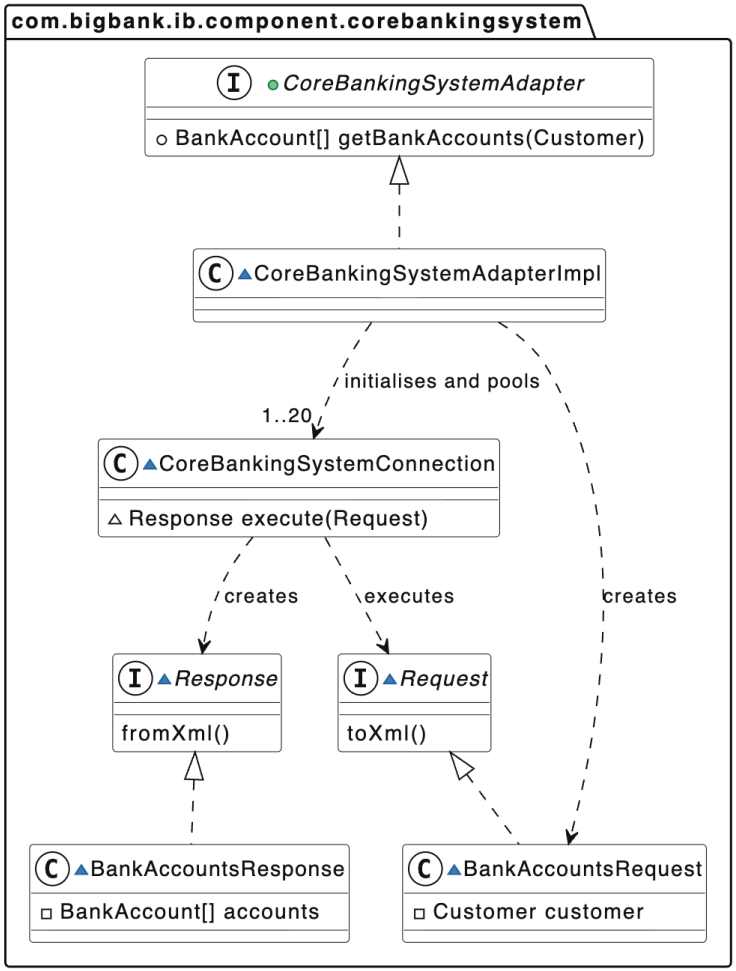
Component Diagram



Component View: Internet Banking System - Backend

The component diagram for the Internet Banking System Backend | Simon Brown | c4model.com | License: CC BY 4.0

Code Diagram



Code View: Internet Banking System - Backend - Core Banking System Adapter

A summary of the implementation details for the Core Banking System Adapter component | Simon Brown | c4model.com | License: CC BY 4.0

End of the Session

